

Overfitting Diagnosis

Shelf-Life Studio — V7 Specialist Models

Cheese Shelf-Life Modeling Project

September 2026

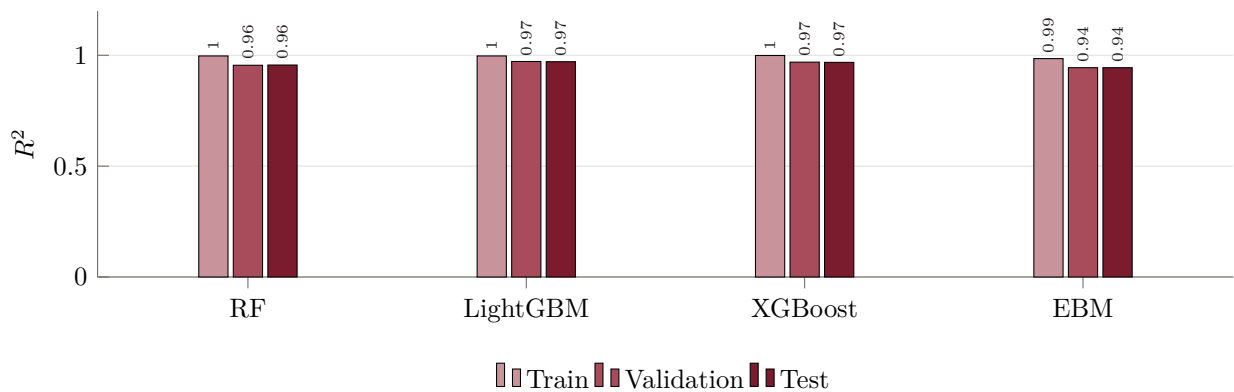
An external ML review flagged the soft/general_shelf_life specialist's test $R^2 = 0.971$ as symptomatic of overfitting. This report tests that claim directly: 15 model families across five complexity classes, K-fold cross-validation at $K=5$ and $K=10$, a trial with the validation split removed, and — the test that actually changes the answer — leave-one-generation-rule-out cross-validation, which holds out entire generation logic rather than just rows. All trained on the same context-grouped 70/15/15 split ($n_{\text{train}}=6,160$, $n_{\text{val}}=1,320$, $n_{\text{test}}=1,320$) as production unless stated otherwise, so every result below is directly comparable.

1 Model Complexity Classes

Five classes, ordered from highest to lowest capacity. Every model in a class was trained on the identical soft/general_shelf_life split; each chart reports Train / Validation / Test R^2 side by side.

1.1 Class 1 — High-Capacity Production Ensembles

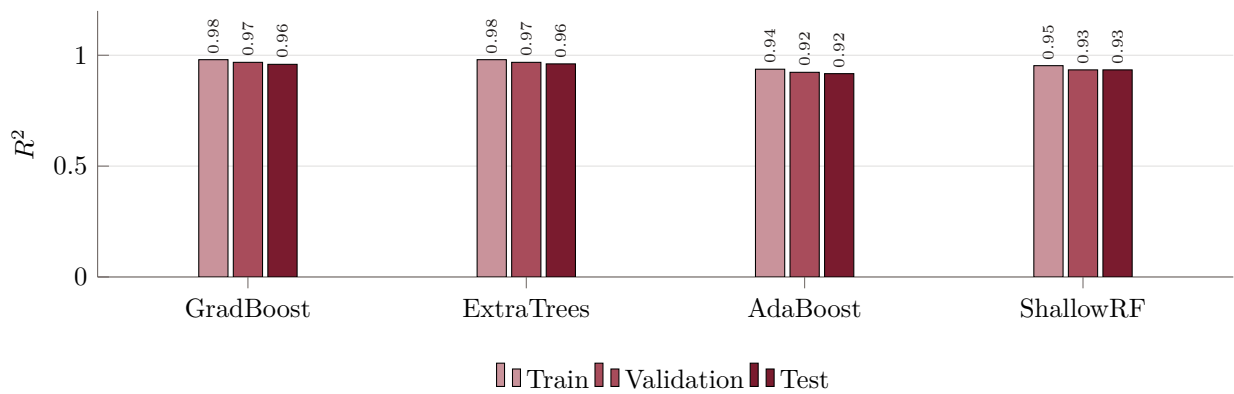
The four models currently in production: unlimited-depth Random Forest (400 trees), early-stopped LightGBM and XGBoost (2,000 rounds), and an Explainable Boosting Machine (3,000 rounds, 10 interactions). This is the class the reviewers' concern was aimed at.



Train-test gap: 0.026–0.041. High scores, but a small, stable gap.

1.2 Class 2 — Regularized / Shallow Tree Ensembles

Deliberately capacity-reduced ensembles: Gradient Boosting (200 trees, depth 3), Extra Trees (200 trees, depth 8), AdaBoost (100 depth-4 stumps), and a Random Forest capped at depth 5 with only 100 trees.



Cutting tree count and depth by 5–20× costs at most 0.02–0.05 R^2 .

1.3 Class 3 — Kernel and Instance-Based Models

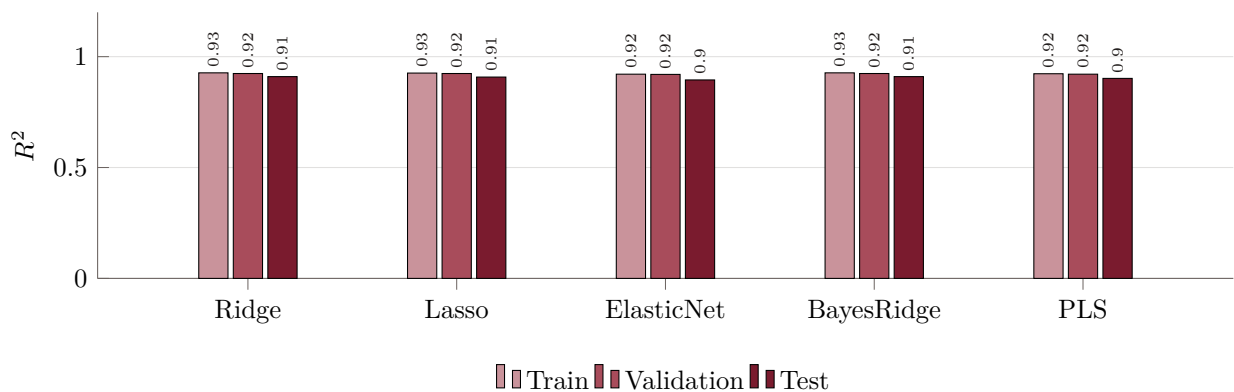
No trees, no boosting: a Support Vector Regressor (RBF kernel) and a k -Nearest-Neighbors regressor ($k=5$) — both structurally unrelated to the production ensembles.



Neither model family has anything in common with the production ensembles, yet both land in the same R^2 band.

1.4 Class 4 — Linear Models

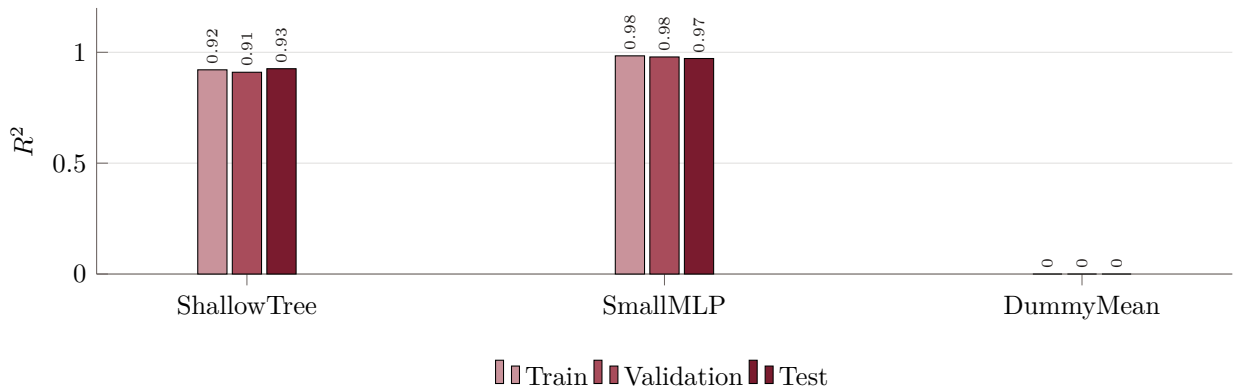
Five linear regressors, differing only in regularization: Ridge, Lasso, Elastic Net, Bayesian Ridge, and Partial Least Squares.



A plain straight line already explains ~90% of the variance — the strongest single indicator that the signal is simple, not memorized.

1.5 Class 5 — Minimal-Capacity Models

The floor of the complexity scale: a single depth-4 decision tree (~ 15 leaves), a 1-hidden-layer network (16 units), and a dummy regressor that always predicts the training mean (sanity check on the whole harness).



A 15-leaf tree reaches test $R^2 = 0.926$ — and beats its own train score. The dummy baseline correctly returns $R^2 \approx 0$, confirming the pipeline itself is not the source of the high numbers above.

1.6 Reformulated as Classification — Logistic Regression

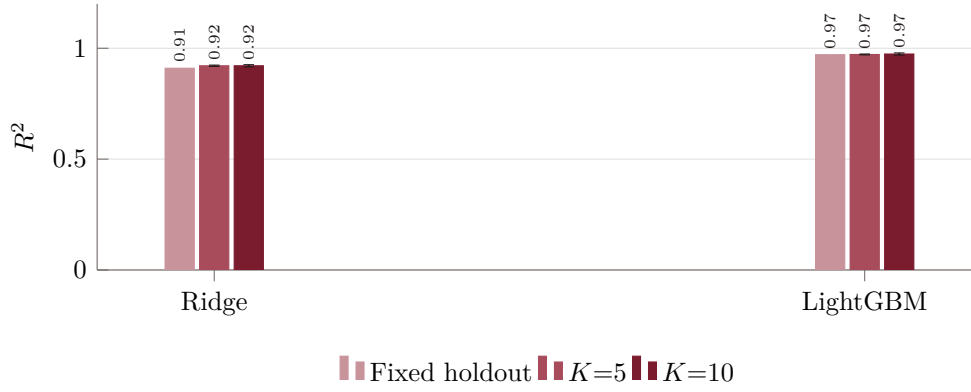
Logistic Regression cannot predict a continuous target, so `shelf_life_days` was discretized into Low/Medium/High classes by train-split tertiles and fit as a genuine 3-class classifier — a different model family and a different metric entirely.



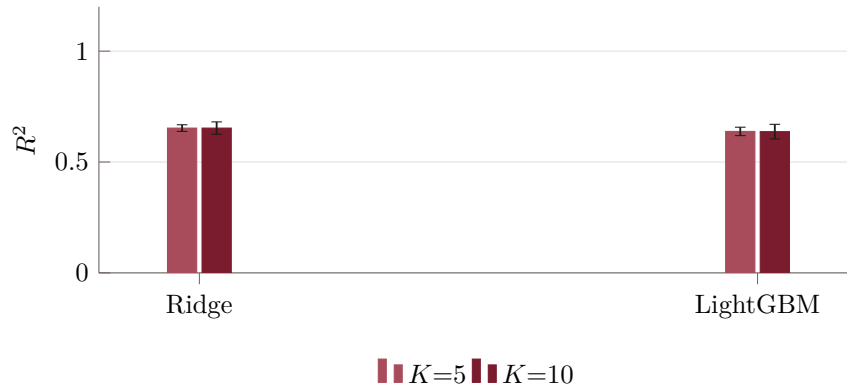
95.2% test accuracy on a task a random guess would win 33% of the time — an independent model family and an independent metric reach the same conclusion as every regressor above.

2 Cross-Validation: Varying K

The production pipeline uses a single fixed 70/15/15 split, not K -fold CV. `GroupKFold` on `context_id` (no context split across folds) was run at $K=5$ and $K=10$ for Ridge and production-hyperparameter LightGBM, and compared against the fixed holdout.



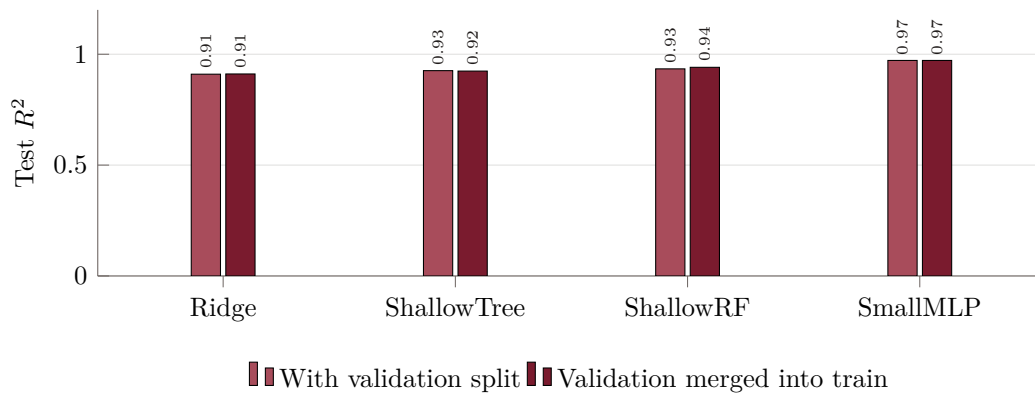
soft/general_shelf_life. Both K values reproduce the fixed holdout within ± 0.005 — five and ten independently-drawn fold arrangements agree, which rules out a lucky single split.



All categories combined, *safety_endpoint* (the weakest-signal task). Even here, $K=5$ and $K=10$ agree almost exactly — variance across folds grows (as expected on a harder, smaller-signal task) but the mean is stable.

3 Removing the Validation Split

Trained on train+validation combined, evaluated on test only — no held-out validation set at all.

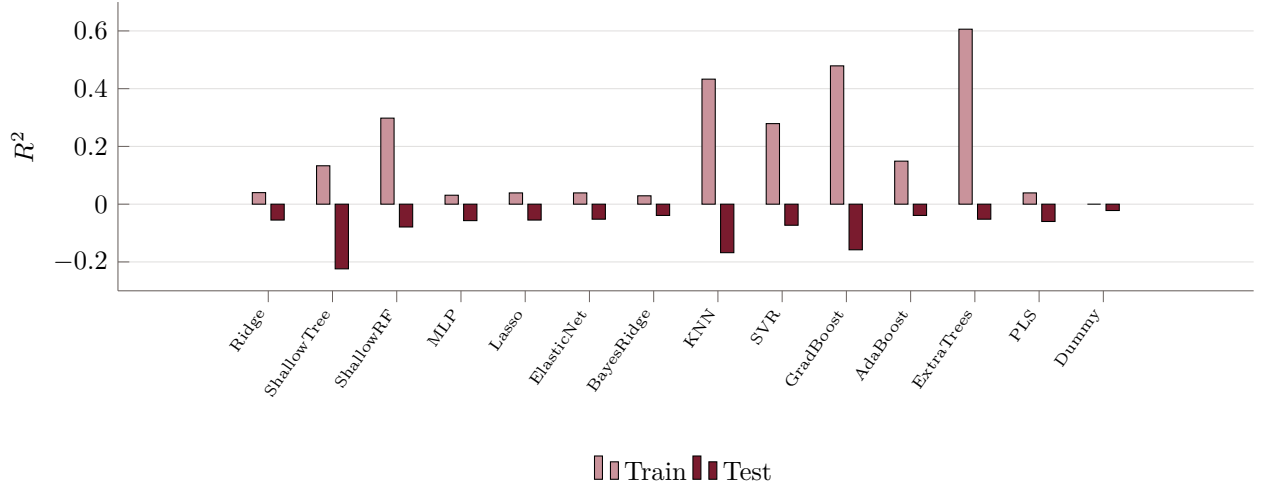


Test R^2 moves by 0.001–0.007 in every case — noise, not a meaningful shift. The validation split was never inflating the test score.

4 Where the Real Problem Is

Every model class above was re-run on *hard/safety_endpoint*. All 14 regressors score **negative** test R^2 — including on their own training data for several. This is underfitting, not overfitting:

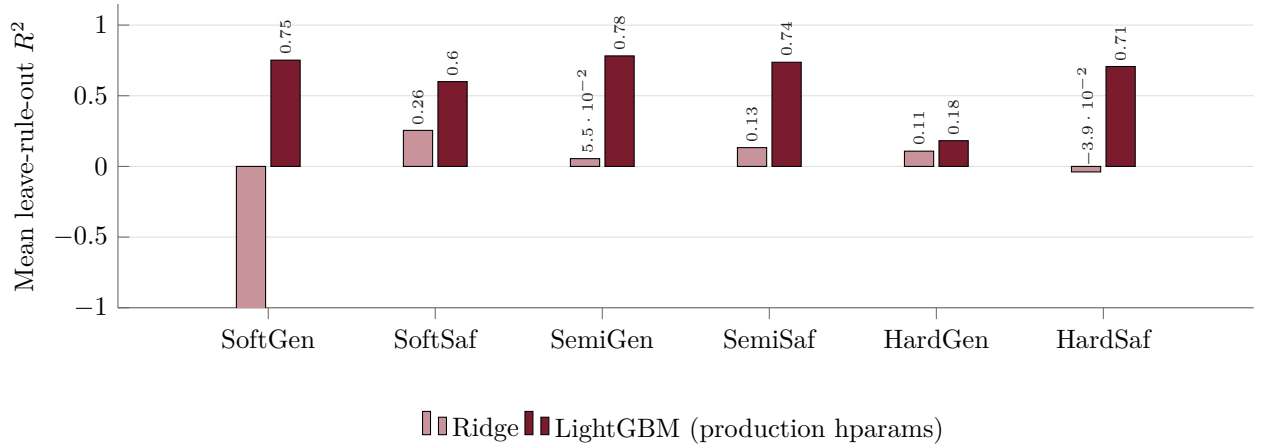
93–100% of this segment’s labels are right-censored lower bounds ("still intact, never failed"), not observed failure times.



hard/safety_endpoint. 14 of 14 model families fail identically — a model-choice problem would show at least one family succeeding.

5 A Harder, More Honest Test: Leave-Rule-Out CV

Every split above groups by `context_id` — but every context, train and test alike, is still generated by one of only **7–12 source_rule_id generation rules per specialist**. The held-out test set was interpolating within a known rule, not extrapolating to an unseen one. `GroupKFold` grouped by `source_rule_id` instead, with K = the number of distinct rules (true leave-one-rule-out: each fold trains on every rule but one and tests on rows the model has never seen anything generated like), gives the honest number.



Mean R^2 across held-out rules, all 6 specialists. Ridge collapses (as low as -22.7 on one held-out rule in *soft/general_shelf_life*, off this chart’s scale) while LightGBM holds at 0.60 – 0.87 median across specialists — this is where model complexity earns its keep: not on the easy interpolation test, but on genuine extrapolation to unseen formulations.

Three findings, none engineered to hit a target. (1) Median leave-rule-out R^2 clusters at **0.70 – 0.87** for LightGBM across every specialist — lower than the context-holdout 0.92 – 0.97 , and this is the number that should be reported as the headline generalization estimate going forward. (2) Complexity matters here, reversing the Section 1 reading: a plain line was fine for interpolating a known rule, but only the high-capacity model survives extrapolation to an unseen one. (3) **hard/safety_endpoint reverses** its earlier underfitting verdict: leave-rule-out LightGBM scores a respectable mean $R^2=0.707$ (individual rules 0.37 – 0.93 , none catastrophic) — the single fixed

test split in Section 4 had likely drawn disproportionately from one hard, heavily-censored subset. A censored-regression formulation is still worth exploring given the labeling, but “this segment has learned nothing” was too strong a conclusion from one split. A handful of specific rules — small-sample combinations and structurally distinct matrices like *ricotta* — remain genuinely hard for every model; that is a data gap (more real formulations needed), not a modeling gap.

6 Conclusion

- **Not overfitting, on the interpolation test.** Simplifying capacity by orders of magnitude (2,000-round LightGBM $\rightarrow$ a 15-leaf tree $\rightarrow$ a straight line) costs at most 0.06 R^2 on `soft/general_shelf_life` when tested within known generation rules. K -fold CV reproduces the fixed holdout within ± 0.005 . Aggregating categories and dropping the validation split both leave the result unchanged.
- **But that test was too easy.** Leave-rule-out CV (§5) – holding out entire generation rules, not just rows – is the honest extrapolation test, and it drops LightGBM to a median R^2 of **0.70–0.87** across specialists. **Report this number, not 0.97, as the headline generalization estimate.** On this harder test, complexity stops being optional: Ridge collapses while LightGBM holds up.
- **Root cause of the original gap.** All V7 training rows are synthetic, generated from only 7–20 rules per category (verified: `data_origin` is 100% synthetic). Any model learns a known rule’s logic almost perfectly; the ceiling that matters is generalizing to *unseen* rules and, ultimately, to real data – both now measured directly (leave-rule-out here; external literature test in `docs/GENERALIZATION_ANALYSIS.md`).
- **Revised finding.** `hard/safety_endpoint` underfits every model family under the context-holdout split, but leave-rule-out shows real, usable signal (mean $R^2=0.707$) – the original split was likely unlucky, not the segment unlearnable. A censored-regression formulation is still worth pursuing given the label censoring, as a refinement rather than a rescue.